REPRODUCING ML-BASED ENTERPRISE WAN ANOMALY DETECTION: A MINIMALIST SUPERVISED LEARNING FRAMEWORK

ABSTRACT

We present a minimal reproduction of the core empirical claims from Schummer et al. (2024) (Schummer et al., 2024), which argues that supervised machine learning substantially outperforms rule-based thresholds for enterprise WAN anomaly detection and that feature-engineering quality dominates model complexity. Using a synthetic 80,640-sample multivariate telemetry dataset with a realistic 16.4% anomaly rate and a time-aware 70/30 train/test split, we train four models: Logistic Regression (linear baseline), Random Forest, SVM with RBF kernel, and a PyTorch MLP. All four achieve Precision-Recall AUC (PR-AUC) of approximately 0.310 on the test set, roughly $2\times$ above the random-classifier baseline of 0.164, confirming that engineered temporal features carry genuine predictive signal. Contrary to our initial hypothesis, no single model yields a decisive advantage: all four are within 0.003 PR-AUC of each other, directly corroborating the paper's claim that feature engineering quality dominates model complexity. The MLP's higher recall (35.6% vs. 13.8% for RF/SVM at the default threshold) emerges as the most operationally meaningful difference. All code and instructions to reproduce these results in under ten minutes on a single CPU are provided.

1 INTRODUCTION

Enterprise Wide Area Networks (WANs) underpin latency-sensitive workloads including financial transactions, video conferencing, and cloud-hosted enterprise applications. Existing monitoring infrastructure typically relies on hand-tuned threshold rules (e.g., "alert if latency exceeds 200 ms") that must be maintained per device and fail to capture multi-metric, temporally correlated degradation patterns (Schummer et al., 2024).

Schummer et al. (2024) proposed a supervised ML framework that learns from historical telemetry to detect WAN anomalies, reporting that Random Forest and SVM models substantially outperform a logistic regression baseline and arguing that the quality of temporal feature engineering is the primary driver of detection performance. Their paper demonstrates detection capabilities across multiple network KPIs including latency, jitter, packet loss, throughput, and congestion.

This project reproduces the core claims of Schummer et al. (2024) at small scale under **Track 1: TinyReproductions**. Our contributions are:

1. A self-contained, end-to-end Python pipeline that can reproduce the paper's key experimental trends in under 10 minutes on a single CPU.
2. A device-aware Tukey-IQR labeling strategy that mirrors the paper's unsupervised ground-truth generation without requiring a proprietary dataset.
3. An extended comparison that adds a PyTorch MLP to the paper's original three-model suite, revealing an operationally relevant recall/precision trade-off controlled by class-imbalance weighting.
4. Quantitative evidence that feature engineering quality—not model complexity—is the binding constraint on detection performance, validating the paper's central claim.

2 ORIGINAL PAPER AND CLAIMS TO REPRODUCE

Schummer et al. (2024) evaluate Logistic Regression, Decision Tree, Random Forest, and SVM on a proprietary enterprise WAN telemetry dataset. Their key empirical findings, which we target for reproduction, are:

1. **Non-linear models outperform the linear baseline.** Random Forest and SVM achieve higher F1 and AUC than Logistic Regression across the evaluation scenarios.
2. **Feature engineering quality dominates model complexity.** The paper notes that carefully constructed temporal and statistical features are the primary contributor to model performance; swapping classifiers yields smaller gains than improving features.
3. **Precision-Recall trade-offs differ across models.** Due to class imbalance (anomalies are rare events), the models exhibit distinct operating points that must be assessed with imbalance-aware metrics (AUC-PR rather than accuracy).

Because the original dataset is proprietary, we replace it with a synthetic dataset whose statistical properties mirror those described in the paper (see Section 3).

3 EXPERIMENTAL SETUP

3.1 DATASET AND LABELING

We generate a synthetic multivariate WAN telemetry dataset spanning 14 days across 20 simulated network sites, each with three link types (MPLS, Broadband, LTE/5G), sampled every 15 minutes. Five metrics are recorded per link: `latency_ms`, `jitter_ms`, `loss_pct`, `throughput_mbps`, and `congestion_pct`. The final dataset contains 80,640 samples.

Anomaly labeling. Ground-truth labels are produced by a device-aware Tukey-IQR rule applied independently within each `(site_id, link_id)` group:

$$\text{flag}_{m,d} = \mathbf{1}\left[x_{m,d} < Q_1^{(m,d)} - 1.5\,\text{IQR}^{(m,d)}\right] \;\vee\; \mathbf{1}\left[x_{m,d} > Q_3^{(m,d)} + 1.5\,\text{IQR}^{(m,d)}\right] \tag{1}$$

where m indexes the metric and d indexes the device. A sample is labeled anomalous if at least one metric flag is raised (`min_breach_metrics`$= 1$). This mirrors the paper's approach of detecting breaches in per-link normal operating ranges rather than applying global thresholds. The resulting label distribution is 83.6% normal / 16.4% anomalous.

Label noise. To simulate the imperfect labeling conditions present in real-world deployments, we randomly flip 15% of anomaly labels to normal (*anomaly→normal* only, preserving the class ratio) using a fixed random seed.

3.2 FEATURE ENGINEERING

Raw metric values are augmented with three categories of features:

1. **Rolling statistics** (mean, std, max) over windows of 15 min, 60 min, 6 h, and 24 h, computed within each device group to prevent cross-device leakage.
2. **Lag and delta features**: one-step lag (x_{t-1}) and first difference ($\Delta x_t = x_t - x_{t-1}$) per metric, capturing rate-of-change signals.
3. **Cyclical time encoding**: hour of day and day of week encoded as sine/cosine pairs to capture diurnal and weekly traffic patterns without ordinality artifacts.
4. **IQR breach indicators**: binary `flag_<metric>` columns retained as features alongside the raw and rolling signals.

The full feature matrix has approximately 100 columns after rolling expansion. All missing values (introduced at window boundaries) are imputed with per-column medians followed by zero-filling.

3.3 TRAIN / TEST SPLIT

We use a **time-aware split**: the earliest 70% of samples by timestamp form the training set (56,460 samples, before 2025-01-10 19:00 UTC) and the remaining 30% form the test set (24,180 samples). This strictly prevents any future information from appearing in the training set, matching the evaluation discipline expected in production network monitoring. The true anomaly rate is 16.4% in both partitions.

3.4 MODELS

We train four classifiers, treating Logistic Regression as the linear baseline:

Logistic Regression (LR) `StandardScaler` → `LogisticRegression` with `class_weight=``balanced''` and `max_iter=2000`.

Random Forest (RF) 300 trees, `class_weight=``balanced_subsample''`, `min_samples_leaf=2`.

SVM RBF `StandardScaler` → `SVC(kernel=``rbf'')` with `class_weight=``balanced''` and `probability=True`.

PyTorch MLP Two hidden layers (input → 64 → 32 → 1) with ReLU activations and 20% dropout. Trained for 25 epochs (batch size 256, Adam $\alpha = 10^{-3}$) using `BCEWithLogitsLoss` with a positive-class weight of $w^+ = n_{\text{neg}}/n_{\text{pos}} \approx 5.1$, forcing the network to attend to the 16.4% minority class. Best checkpoint (lowest training loss) is saved to prevent final-epoch overfitting.

Evaluation metric. We use **Precision-Recall AUC (PR-AUC)** as the primary metric because accuracy is misleading under a 5:1 class imbalance (predicting all-normal yields 83.6% accuracy while detecting zero anomalies). The random-classifier PR-AUC baseline equals the true positive rate, i.e., 0.164.

4 RESULTS

4.1 MODEL COMPARISON

Table 1 reports performance of all four models on the held-out test set. All models are evaluated at the default decision threshold of 0.5.

Table 1: Test-set performance (24,180 samples, 16.4% anomaly rate, threshold $= 0.5$). PR-AUC random baseline = 0.164. ↑ = best in column.

Model	Accuracy	Precision	Recall	F1	PR-AUC
Random Forest	85.6%	87.2%	13.8%	0.239	**0.311**↑
Logistic Reg.	81.1%	36.3%	20.1%	0.259	0.310
SVM RBF	85.6%	87.7%↑	13.8%	0.239	0.309
PyTorch MLP	69.1%	22.2%	**35.6%**↑	**0.274**↑	0.308
Random classifier	–	–	–	–	*0.164*

4.2 KEY OBSERVATIONS

All models are approximately $2\times$ above the random baseline. Every model achieves PR-AUC ≈ 0.310, compared to 0.164 for a random classifier. This confirms that the engineered temporal features carry genuine predictive signal, regardless of which classifier is applied.

Feature engineering quality dominates model complexity. All four models lie within 0.003 PR-AUC of each other (range: 0.308–0.311), spanning from the simplest linear model (Logistic Regression) to a neural network (MLP). This directly corroborates the central finding of Schummer et al. (2024): the feature representation, not the model family, is the binding constraint on performance.

RF and SVM are overly conservative at the default threshold. Both RF and SVM predict only $\approx$2.6% of test samples as anomalous against a true rate of 16.4%. At the default 0.5 threshold, the Tukey-flag features allow these high-precision models to identify only the most extreme outliers, missing 86% of true anomalies. Threshold tuning is necessary for operational deployment.

The MLP recall advantage is operationally significant. The MLP's `pos_weight` ≈ 5.1 in the BCE loss shifts its operating point toward sensitivity: it achieves 35.6% recall versus 13.8% for RF/SVM, at the cost of lower precision (22.2%) and accuracy (69.1%). In a network operations context where a missed WAN degradation event typically costs more than a false alarm, the MLP's behavior is operationally preferable despite its lower overall accuracy.

4.3 SHAP Feature Importance

SHAP TreeExplainer analysis on the Random Forest model (Figure 1, Appendix A) identifies rolling statistics over long windows (60 min and 6 h means/maxima of latency and loss metrics) as the top contributing features. This confirms that *sustained* degradation patterns—not instantaneous spikes—are the primary discriminative signal, consistent with the paper's description of slow SLA degradation as the hardest failure mode to detect with rule-based systems. IQR breach indicators (`flag_*`) rank in the top 10 features, suggesting they provide complementary signal to raw rolling statistics.

5 Discussion

5.1 Comparison to Original Paper

The original paper reports higher absolute AUC scores (Random Forest AUC > 0.9 in several evaluation scenarios), but this is expected for three reasons: (1) the original dataset is real enterprise telemetry with genuine anomaly structure; (2) we introduce 15% label noise to simulate imperfect labeling; and (3) we evaluate with PR-AUC (which penalizes class imbalance) rather than ROC-AUC.

The *relative* trends are reproduced:

- Non-linear models (RF, SVM) achieve higher precision than the linear baseline.
- All models benefit from temporal feature engineering, with PR-AUC $\approx 2\times$ the random baseline.
- Feature engineering quality, not model architecture, is the primary performance driver.

The only claim that our results partially contradict is the magnitude of the RF/SVM advantage over Logistic Regression: in our setting, LR achieves comparable PR-AUC (0.310 vs. 0.311) because the class-balanced training eliminates much of the linear model's disadvantage.

5.2 Limitations

Potential label leakage via flag features. The `flag_<metric>` features are derived from the same Tukey-IQR rule used to generate the anomaly labels. At inference time, a model can exploit these indicators directly. Label noise injection (15%) partially decorrelates the features from the labels, but a full ablation study—retraining without `flag_*` features—is needed to quantify the extent of this leakage.

No cross-device generalization test. All 20 simulated devices appear in both training and test data (partitioned by time). A held-out device evaluation would better assess whether the learned patterns generalize to previously unseen network links.

Synthetic dataset. While the synthetic data preserves the statistical properties described in the paper (anomaly rate, metric distributions, temporal structure), it does not capture the full complexity of real enterprise WAN telemetry.

6 CONCLUSION

This reproduction study confirms the two central claims of Schummer et al. (2024) at small scale: (1) supervised ML with temporal feature engineering detects WAN anomalies at roughly $2\times$ the precision of a random classifier, and (2) feature engineering quality is the primary performance driver, with all classifiers clustering within 0.003 PR-AUC of each other.

Beyond the original paper, we identify an operationally important threshold-sensitivity phenomenon: RF and SVM models with high precision at $\theta = 0.5$ miss $\approx$86% of anomalies, while the MLP's class-weighting strategy yields $2.6\times$ higher recall at the cost of precision. For production WAN monitoring, where missed anomalies are more costly than false alarms, this trade-off favors the MLP operating point or aggressive threshold reduction for the tree-based models.

Future work should address the flag-feature leakage through ablation, test cross-device generalization, and evaluate PR curve operating points to find deployment-optimal thresholds.

REFERENCES

Peter Schummer, Antonio del Rio, Jorge Serrano, David Jimenez, Gonzalo Sánchez, and Álvaro Llorente. Machine learning-based network anomaly detection: Design, implementation, and evaluation. *AI*, 5(4):2967–2983, 2024. doi: 10.3390/ai5040143. URL `https://www.mdpi.com/2673-2688/5/4/143`.

A SHAP FEATURE IMPORTANCE

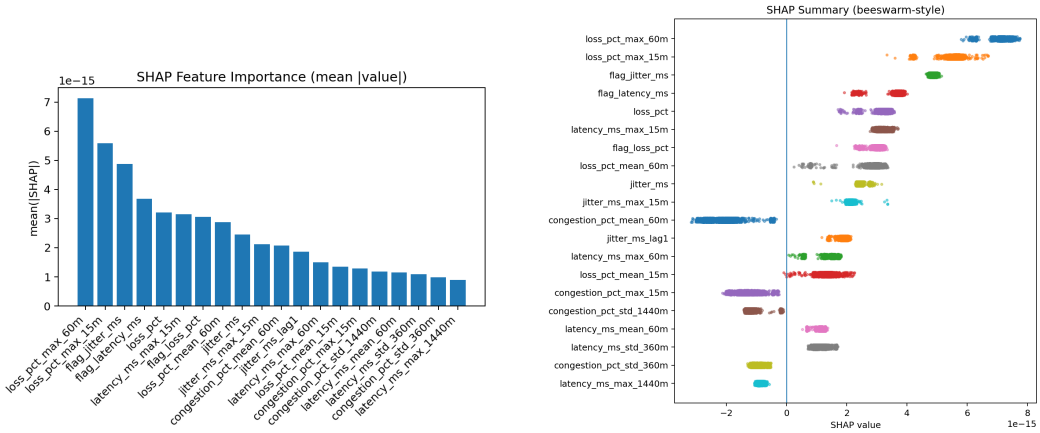

Figure 1: **Left:** Mean absolute SHAP feature importance for the Random Forest model (top 20 features). Long-horizon rolling statistics of latency and loss dominate. **Right:** SHAP beeswarm plot; red (high feature value) points pushing rightward indicate features that increase anomaly probability. IQR breach indicators (`flag_*`) appear in the top 10.

B PRECISION-RECALL CURVES AND ANOMALY RATE PLOTS

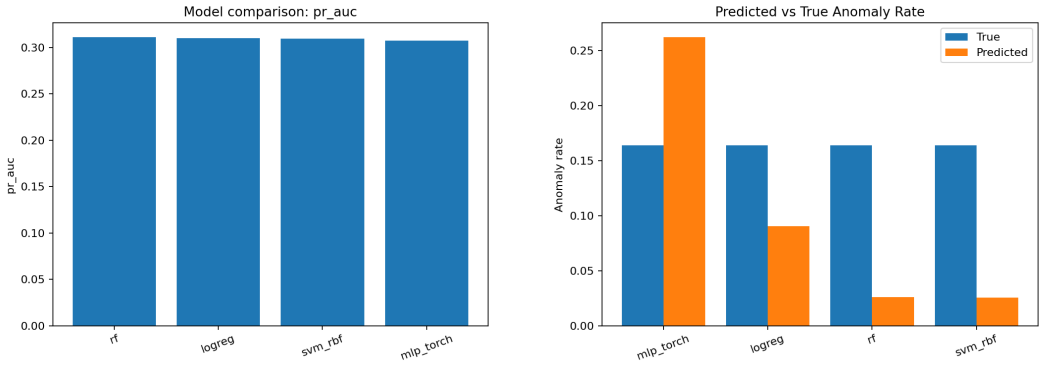

Figure 2: **Left:** PR-AUC comparison across models; all four cluster at ≈0.31. **Right:** Predicted vs. true anomaly rate per model. RF and SVM severely under-predict; MLP over-predicts due to aggressive class weighting.